

PHILADELPHIA INTL, PA, USA

WMO: 724080

Lat: 39.873N

Lon: 75.227W

Elev: 10

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13739

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.8	17.8	-5.2	4.2	16.8	-0.9	5.2	20.9	28.6	34.0	26.4	34.8	12.6	290	0.509

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.1	93.4	75.0	90.8	73.9	88.4	72.5	78.1	88.3	76.7	85.8	75.5	83.6	11.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.3	132.6	82.5	74.0	126.9	81.1	72.9	122.0	80.2	41.5	88.5	40.0	85.6	38.8	84.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.8	21.2	18.9	8.7	97.5	5.4	2.9	4.8	99.6	1.7	101.3	-1.3	102.9	-5.2	105.0
			6.3	80.5	4.9	2.1	2.8	82.0	-0.1	83.2	-2.9	84.4	-6.5	85.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.8	33.8	36.2	44.0	55.1	64.6	73.8	78.9	77.2	70.4	58.8	47.8	39.1
	DBStd	17.46	9.96	9.12	8.92	8.16	7.33	6.15	4.87	4.83	6.28	7.64	8.11	8.60
	HDD50	1672	511	394	230	37	0	0	0	0	0	12	135	353
	HDD65	4410	968	806	653	313	100	8	0	0	21	222	517	802
	CDD50	4136	8	8	43	190	452	714	896	842	613	285	69	16
	CDD65	1403	0	0	1	17	87	272	431	378	185	31	1	0
	CDH74	12217	0	0	15	177	761	2355	4248	3284	1211	164	2	0
	CDH80	4471	0	0	2	48	235	851	1780	1196	334	25	0	0
Wind	WSAvg	9.1	10.1	10.2	10.7	10.2	8.8	8.5	8.2	7.6	8.0	8.7	9.2	9.5
Precipitation	PrecAvg	45.9	3.7	2.7	4.2	3.7	3.6	4.1	4.5	4.8	3.8	3.6	3.2	4.0
	PrecMax	59.1	5.8	5.4	7.7	9.1	7.1	9.4	10.3	14.8	10.4	9.3	6.1	9.0
	PrecMin	34.1	1.6	0.4	1.1	1.1	1.5	1.3	1.6	1.6	0.2	0.6	1.0	0.7
	PrecStd	6.6	1.2	1.3	1.7	1.7	1.3	2.2	2.0	2.7	2.4	2.0	1.3	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.4	66.6	75.5	85.9	90.3	94.5	97.4	95.7	91.5	83.2	72.0	67.2
		MCWB	59.3	56.8	60.6	66.1	71.6	74.9	77.7	76.5	73.2	69.9	62.3	60.0
	2%	DB	58.9	60.0	68.3	79.2	86.1	91.2	94.2	91.9	87.3	78.6	68.0	61.2
		MCWB	55.1	51.8	56.6	62.6	69.4	73.6	75.4	75.0	71.4	67.4	60.5	56.2
	5%	DB	52.9	54.4	62.9	73.7	82.2	88.0	91.4	88.9	83.7	74.6	64.2	56.5
		MCWB	48.2	47.0	53.3	60.2	67.6	72.1	74.5	73.2	70.2	65.2	57.5	52.1
	10%	DB	47.6	49.5	58.1	69.3	78.1	85.0	88.7	86.4	80.8	71.0	60.5	51.8
		MCWB	42.8	43.0	50.1	57.4	65.5	70.4	73.3	71.7	68.6	62.9	54.3	46.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.9	60.5	63.4	68.7	73.8	77.9	80.3	79.7	76.9	73.1	65.7	62.4
		MCDB	63.2	65.2	72.4	80.1	85.6	89.4	92.1	90.0	84.5	78.0	68.9	65.8
	2%	WB	55.7	53.5	58.8	65.6	71.8	76.1	78.4	77.9	75.2	70.2	62.4	57.3
		MCDB	59.0	57.6	65.2	75.2	82.5	86.2	89.1	87.3	81.8	76.0	66.6	60.4
	5%	WB	48.8	48.1	55.0	62.7	69.7	74.8	77.1	76.4	73.6	67.5	58.9	52.9
		MCDB	52.0	52.7	61.1	71.3	79.1	84.0	86.9	84.4	79.2	72.5	63.0	55.8
	10%	WB	43.9	43.8	50.8	59.5	67.7	73.2	75.7	75.1	71.8	64.4	55.6	47.9
		MCDB	46.8	48.7	57.7	67.1	75.7	81.5	84.4	82.1	77.9	69.7	59.4	51.1
Mean Daily Temperature Range	5% DB	MDBR	13.2	14.6	16.6	18.4	17.8	16.9	16.1	15.5	15.8	15.6	14.8	12.8
		MCDBR	19.8	22.0	24.3	24.9	22.7	20.1	18.8	17.8	18.6	19.3	19.5	18.3
		MCWBR	16.7	16.4	15.5	12.8	10.1	7.9	6.5	6.6	7.7	10.2	13.3	16.1
	5% WB	MCDBR	19.1	20.5	23.1	22.4	20.6	18.2	17.1	15.8	15.2	16.1	16.4	17.6
MCWBR		17.6	17.3	16.9	13.9	10.7	8.4	6.7	7.0	8.5	11.1	13.9	17.0	
Clear-Sky Solar Irradiance	taub	0.317	0.323	0.366	0.398	0.444	0.507	0.512	0.504	0.416	0.363	0.329	0.312	
	taud	2.444	2.419	2.357	2.296	2.198	2.066	2.065	2.074	2.294	2.447	2.505	2.509	
	Ebn at Noon	269	284	281	278	266	249	246	244	261	266	263	263	
	Edh at Noon	26	30	36	40	46	52	52	50	38	29	24	23	
All-Sky Solar Radiation	RadAvg	614	901	1189	1539	1716	1878	1909	1666	1377	987	702	534	
	RadStd	53	100	95	131	206	149	89	106	150	102	62	55	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.70	N/A	N/A	N/A	N/A	N/A	N/A	+196	+140
(47) Regional (75 neighbors)		+0.76	N/A	N/A	N/A	N/A	N/A	N/A	+225	+131

Nomenclature: See separate page